VitalEye: A Comprehensive Multi-Model Framework for Robust Remote Heart Rate and Respiratory Rate Monitoring

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GitHub: https://github.com/rpyaduvanshi950/VitalEye-Cs776
Deployment: https://huggingface.co/spaces/pushpender-23/VitalEye_CS776

***Abstract*—This project presents VitalEye, a contactless vital sign monitoring system for estimating heart rate (HR) and respiratory rate (RR) from facial video data. Traditional monitoring approaches rely on contact-based sensors such as ECG and respiratory belts, which require physical attachment, leading to discomfort, restricted mobility, and limited scalability for continuous or remote healthcare applications.**

To address these limitations, we propose a modular pipeline integrating computer vision, preprocessing, deep learning, and signal processing. The system processes video input captured using a standard RGB camera and detects key regions of interest (ROI), including the face and chest, using an object detection model. Preprocessing techniques such as motion stabilization, spatial masking, and temporal filtering are applied to reduce noise and improve signal quality.

The refined signals are then passed through a spatiotemporal deep learning model to recover physiological waveforms. These signals are further analyzed in the frequency domain using Fast Fourier Transform (FFT) to identify dominant periodic components corresponding to heart rate and respiratory rate.

By combining ROI-based detection, robust preprocessing, and deep learning-based signal extraction, the proposed system enables real-time, non-contact estimation of vital signs using standard video input, making it suitable for scalable remote healthcare applications.

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Monitoring vital signs such as heart rate (HR) and respiratory rate (RR) is essential for assessing a patient's physiological condition in both clinical and remote healthcare environments. Conventional monitoring systems rely on contact-based sensors, including ECG devices and respiratory belts, which require direct attachment to the patient. While accurate, these methods can cause discomfort, restrict movement, and limit long-term continuous monitoring.

Recent advancements in computer vision have enabled contactless physiological monitoring using video-based approaches. Subtle variations in facial skin color caused by blood flow and chest movements associated with respiration can be captured using standard RGB cameras and analyzed over time to estimate vital signs.

However, extracting these signals remains challenging. The physiological signals are extremely weak and are highly sensitive to motion, illumination changes, and environmental noise. Even minor head movements or lighting variations can distort the signal significantly.

To address these challenges, this project proposes a structured end-to-end pipeline consisting of:

- Video acquisition using RGB cameras
- ROI detection (face and chest regions) using object detection models
- Preprocessing including motion stabilization and filtering
- Deep learning-based signal extraction using spatiotemporal models
- Frequency-domain analysis using FFT for HR and RR estimation

This integrated approach enables a robust, real-time, and scalable system for contactless vital sign monitoring.

II. Problem Statement

The primary problem addressed in this project is the accurate extraction of physiological signals (HR and RR) from video data without using physical sensors.

While contactless methods based on video analysis offer significant advantages, they face several critical challenges in real-world deployment:

A. Key Challenges

1) **Motion Artifacts**: Subject movements such as head motion or body shifts introduce large variations that overshadow subtle physiological signals.
2) **Illumination Variations**: Changes in lighting conditions affect pixel intensity and introduce noise, making signal extraction unreliable.
3) **Weak Signal Strength**: Physiological signals such as pulse-induced color changes and breathing motion are extremely subtle compared to background variations and sensor noise.
4) **Region Localization**: Accurate detection of relevant regions (face and chest) is essential; incorrect ROI selection leads to poor signal quality.

B. Objective

The objective of VitalEye is to develop a robust, real-time contactless monitoring system that:

- Accurately estimates heart rate (HR) and respiratory rate (RR)
- Works under motion and illumination variations
- Uses standard RGB video input without physical sensors
- Leverages ROI detection, preprocessing, deep learning, and frequency analysis

The system aims to provide a scalable and practical alternative to traditional contact-based monitoring by integrating detection and signal extraction into a unified pipeline.

III. Architectural Overview

The VitalEye system is organized into a modular pipeline that processes raw video input and yields clinical-grade physiological metrics. This unified pipeline integrates detection, deep learning, and signal processing into a seamless flow.

High-Level Pipeline Components: 1. Video Input: Capturing the subject's face at 30 FPS or higher using a standard RGB camera. 2. ROI Detection: Using YOLOv8 to identify the Face, Nose, and Chest regions for targeted analysis. 3. Preprocessing: Applying stabilization, spatial masking, and temporal filtering to clean the signal. 4. Signal Extraction Model: Feeding the cleaned tensors into a spatiotemporal deep learning model (e.g., PhysFormerX). 5. Frequency Analysis: Converting the predicted 1D waveform into the frequency domain via FFT for peak identification. 6. Final Output: Displaying the estimated HR in BPM and RR in BrPM to the user.

By separating the detection and extraction phases, the system achieves high efficiency and accuracy. The detection layer ensures that only relevant pixels are processed, while the deep learning layer focuses on the complex task of signal recovery from the stabilized video.

A. YOLOv8 Detection Pipeline

After preparing the dataset from the UBFC-rPPG, a custom object detection model was trained using YOLOv8 to identify key regions of interest (ROIs) relevant for physiological signal extraction. The input to the model consisted of extracted video frames along with corresponding bounding box annotations in standard YOLO format, covering three classes: Face, Nose, and Chest. The dataset was organized in a structured directory compatible with YOLO training pipelines, ensuring efficient data loading and scalability.

During training, the model learned to perform multi-class detection by optimizing bounding box localization and classification accuracy simultaneously. To improve generalization and robustness, extensive data augmentation techniques were applied, including random rotations, horizontal flipping, scaling, cropping, and noise injection. These augmentations enabled the model to handle variations in lighting conditions, subject motion, and partial occlusions, which are common in real-world scenarios.

Once trained, the YOLOv8 model was capable of accurately predicting three bounding boxes per frame corresponding to the Face, Nose, and Chest regions. The Face region serves as the primary source for heart rate estimation, the Nose provides a stable and highly perfused area for refining the physiological signal, and the Chest region is used for respiratory rate estimation. The trained model was then deployed on the original (uncompressed) videos to extract ROI-specific sub-videos, resulting in a structured dataset where each video contains separated Face, Nose, and Chest clips along with synchronized ground truth signals.

The choice of YOLOv8 was motivated by its anchor-free detection mechanism, strong speed–accuracy trade-off, and lightweight architecture, which allows real-time performance even on CPU-based systems. Its robustness in maintaining stable detections under motion and partial occlusion makes it particularly suitable as the foundational step in the rPPG pipeline.

Fig. 1. Visualization of YOLOv8-based ROI detection, showing accurate bounding box localization for Face, Nose, and Chest regions across varying subjects and environments.

B. ROI Stabilization & Skin Extraction

This stage represents the core engineering of the VitalEye system. While many projects focus solely on the machine learning model, we found that the quality of the input data is the primary driver of accuracy. We implemented several critical techniques: 1. Face Stabilization via EMA: Even with a high-quality detector, bounding boxes often exhibit slight 'jitter' between frames. We implement an Exponential Moving Average (EMA) filter on the box coordinates: $Box(t) = 0.1 \times RawBox(t) + 0.9 \times Box(t-1)$. This smoothing removes the high-frequency jitter that would otherwise be interpreted as pulse noise by the model. 2. Skin Segmentation in YCbCr Space: We convert the frames to the YCbCr color space. Human skin clusters in a very specific region of the Cb and Cr chrominance planes. By applying thresholding followed by morphological opening and closing, we isolate the skin pixels of the forehead and cheeks while removing eyes, hair, and the background. This ensures the spatial average only contains pulse info. 3. Illumination Normalization: We apply a 30-frame running mean subtraction to the averaged color signals. This effectively neutralizes slow-moving lighting drifts from environment changes, leaving only the high-frequency physiological signal for analysis.

C. Signal Preprocessing & Ground Truth Alignment

To train a deep learning model for rPPG, the video data must be perfectly synchronized with the physiological ground truth. Ground Truth Processing: The raw PPG signals from the oximeter (often recorded at 1000 Hz) were interpolated to match the exact timestamp of each video frame (typically 30 FPS). This synchronization is vital for models like PhysFormerX that rely on temporal alignment. Filtering: We applied a 2nd-order Butterworth bandpass filter with a passband of 0.7 Hz to 2.5 Hz. This filter isolates the relevant physiological frequencies while suppressing low-frequency motion and high-frequency sensor noise. Data Preparation: The continuous video and label streams were divided into standardized 300-frame windows (approx. 10 seconds). We used a 50% overlapping stride to double our effective dataset size and provide the model with varied temporal starting points. The final output of this stage is a set of PyTorch tensors (.pt) with subject-independent splits to ensure the model generalizes to people it has never seen during training.

D. Input Representation Engineering

One of the strongest technical contributions of this project is the 6-channel input representation. Most rPPG models use simple RGB frames as input. However, we found that providing the model with more explicit temporal information leads to significantly better performance. Key Idea: We create a 6-channel tensor where the first three channels are the original RGB frames and the next three channels are the 'Difference Frames' ($Frame_t - Frame_{t-1}$). The difference frames are normalized by the sum ($Frame_t + Frame_{t-1}$) to eliminate illumination scale effects. This representation effectively removes static features (like the shape of the face or hair) and amplifies the subtle temporal variations caused by the pulse. Final Input Shape: [Batch Size, 6 Channels, 300 Frames, 64 Height, 64 Width]. By explicitly providing the temporal gradient, we reduced the learning burden on the model, allowing it to focus on the periodic patterns rather than finding the skin. This engineering choice improved our heart rate estimation accuracy by approximately 3–5 BPM compared to raw RGB inputs.

IV. Architectural Overview of PhysFormerX

A. System-Level Data Flow

The PhysFormerX pipeline constitutes an end-to-end spatiotemporal learning framework that transforms a raw RGB facial video stream into two simultaneous physiological waveforms: the remote photoplethysmography (rPPG) signal for heart rate (HR) estimation and a concurrent low-frequency respiratory signal for respiratory rate (RR) estimation. The system is organized into four sequential stages: (1) preprocessing and input construction, (2) three-dimensional spatiotemporal feature extraction, (3) temporal signal refinement, and (4) frequency-domain physiological metric estimation. Each stage is discussed in detail below.

B. Input Processing and 6-Channel Difference-Frame Representation

The raw facial video is ingested as a sequence of 300 frames at 30 FPS, spatially resized to 64×64 pixels. Each frame is normalized to the floating-point range [0, 1]. Rather than providing the raw RGB frames directly to the model, PhysFormerX constructs an augmented 6-channel input tensor that encodes explicit temporal gradient information alongside the original appearance signal.

The 6-channel representation is computed as follows. The first three channels carry the normalized RGB intensities of the current frame at time t. The latter three channels carry the inter-frame difference signal: $diff[t] = frame[t] - frame[t-1]$, with the first-frame difference initialized to zero. This decomposition is applied uniformly across all frames without any frequency-selective filtering; the preprocessing stage intentionally preserves the full spectrum of temporal variation in both the RGB and difference channels. No bandpass restriction is imposed at this stage, ensuring that physiological components spanning the full range of human cardiac activity (approximately 0.75–2.5 Hz) and respiratory activity (approximately 0.1–0.5 Hz) are retained in the input representation.

The motivating principle behind this design is straightforward: the inter-frame difference is proportional to the instantaneous rate of change of pixel intensity. Because static scene elements — skin texture, hair, background — contribute identical values across consecutive frames, their contribution to the difference channels is negligible. In contrast, the subtle periodic fluctuations in skin perfusion induced by cardiac and respiratory cycles produce non-zero, structured signals in the difference representation. This input construction therefore functions as a spatial high-pass filter in the temporal domain, amplifying the very signals of interest while suppressing stationary nuisance features, without sacrificing the broad spectral bandwidth that the model requires to learn both high-frequency cardiac and low-frequency respiratory signatures.

The final input tensor has shape [B × 6 × T × H × W], where B is the batch size, T = 300 is the temporal sequence length (or T = 150 for the training-time random crop), H = W = 64 are the spatial dimensions, and 6 denotes the stacked RGB and difference channels.

C. Spatiotemporal Feature Extraction: 3D Residual Backbone with SE Attention

The 6-channel input tensor is passed through a hierarchical 3D convolutional backbone that extracts spatiotemporal features at progressively increasing semantic depth while preserving temporal resolution throughout the network. The backbone comprises a convolutional stem followed by four residual stages.

Stem Block. The stem applies a single 3D convolution with kernel dimensions (1, 5, 5) — a temporally flat, spatially wide kernel — projecting the 6-channel input to a 32-channel feature map. The temporal kernel dimension of 1 is a deliberate design choice: it ensures that the full temporal resolution

of the input is preserved at the stem output, so that every frame's feature representation is available to the downstream residual layers for waveform reconstruction. Spatial resolution is halved by a (1, 2, 2) max-pooling operation following batch normalization and ReLU activation.

Residual Stages. The backbone consists of four stacked ResBlock3D modules organized into two pairs. The first pair (Layer 1 and Layer 2) operates at 64 feature channels. Layer 1 applies a spatial stride of (1, 2, 2) to compress the spatial dimensions while keeping the temporal axis intact; Layer 2 maintains the same resolution with stride 1, providing additional depth for feature refinement. The second pair (Layer 3 and Layer 4) mirrors this structure at 128 channels, with Layer 3 again performing (1, 2, 2) spatial subsampling.

Each ResBlock3D consists of two sequential 3D convolutions (kernel $3\times3\times3$, padding 1) with batch normalization and ReLU non-linearity, connected by a learned identity shortcut that applies a $1\times1\times1$ convolution and batch normalization when the channel dimensionality changes. A mild 3D spatial dropout ($p = 0.05$) is applied at each residual block for regularization.

Squeeze-and-Excitation Channel Attention. Embedded within every residual block, the SEBlock3D module implements channel-wise recalibration via global average pooling across all spatiotemporal dimensions, followed by a two-layer fully connected bottleneck (reduction factor of 8) with sigmoid gating. This mechanism allows the network to dynamically attend to feature channels that carry the strongest physiological signal — empirically, the green channel of the original RGB representation and the corresponding green difference channel — while suppressing noisy contributions from other channels. Crucially, because no frequency restriction has been applied to the input, the SE attention operates over the full spectral content of the feature maps. This allows the model to simultaneously amplify both the cardiac-frequency components relevant to HR estimation and the lower-frequency respiratory components required for RR estimation.

After the four residual stages, an adaptive global average pooling layer collapses the spatial dimensions (H and W) to a single point while leaving the temporal axis intact, yielding a feature tensor of shape [B $\times$ 128 $\times$ T].

D. Temporal Signal Refinement Head

The spatially pooled temporal feature tensor is passed through a dedicated 1D convolutional temporal processing module. This module consists of two sequential 1D convolution layers (kernel size 5, padding 2) with 64 output channels each, batch normalization, and ReLU activations. The 1D convolutions operate exclusively along the temporal axis, implementing a learned temporal smoothing and integration operation that aggregates multi-scale temporal context within a receptive field of approximately 9 frames per layer.

Following temporal processing, a final projection head applies a 1D point-wise convolution (kernel size 1) preceded by a 10% dropout layer, reducing the 64-channel temporal representation to a single channel. The output is a 1D waveform of shape [B $\times$ T], which constitutes the model's predicted rPPG signal. This unified output encodes the full temporal dynamics of the reconstructed physiological waveform — encompassing both the dominant cardiac oscillation and the superimposed low-frequency respiratory modulation — without any architectural separation between the two signal components.

E. Dual-Signal Estimation via Frequency-Domain Analysis

Heart rate and respiratory rate are extracted from the predicted 1D waveform through a unified frequency-domain analysis pipeline. The waveform is first mean-subtracted to remove any DC offset, then multiplied by a Hann window to suppress spectral leakage at the temporal boundaries of the 300-frame clip. A periodogram is computed via a 2048-point FFT, yielding a high-resolution power spectral density estimate.

HR estimation is performed by identifying the dominant spectral peak within the cardiac frequency band, defined as 0.75–2.5 Hz (45–150 BPM). RR estimation is performed analogously by identifying the dominant peak within the respiratory frequency band, defined as 0.1–0.5 Hz (6–30 BrPM). Because the preprocessing stage preserves the full spectral bandwidth of the input, and because the 3D backbone and temporal head impose no explicit frequency constraints on the output waveform, both physiological components are present in the predicted signal. The frequency-domain estimation stage therefore acts as the architectural point of signal disentanglement: a single unified waveform is queried over two distinct, non-overlapping frequency intervals to yield independent estimates of HR and RR.

This frequency-domain approach confers a critical robustness advantage over time-domain peak-counting methods. Transient noise spikes in the predicted waveform manifest as a broadband elevation of the spectral noise floor rather than spurious discrete peaks, leaving the dominant physiological frequency clearly identifiable. The approach is therefore significantly more resilient to motion artifacts and illumination transients than alternatives based on threshold crossings or local maxima detection.

F. Training Objective

The model is trained end-to-end using a composite loss function that jointly optimizes three complementary criteria. The first component is a negative Pearson correlation loss, which penalizes deviation between the temporal dynamics of the predicted and ground-truth waveforms, encouraging shape and phase fidelity. The second is a frequency-domain cross-entropy loss (FreqCELoss) that treats HR bin estimation as a classification problem: the predicted waveform is transformed to the frequency domain and its spectral distribution over the cardiac band is compared against the ground-truth spectral distribution via KL divergence, directly incentivizing accurate

frequency localization. The third is a Smooth-L1 loss for pointwise amplitude stability. The combined objective is:

$$L = L_{Pearson} + 0.5 \cdot L_{FreqCE} + 0.1 \cdot L_{SmoothL1} \quad (1)$$

Training employs the AdamW optimizer with OneCycleLR scheduling (10% warm-up, cosine annealing) and mixed-precision execution. Exponential Moving Average (EMA, decay = 0.998) of model weights is maintained throughout training and applied during validation and inference, significantly improving prediction smoothness. Training is terminated by early stopping with a patience of 15 epochs, monitored on subject-held-out validation BPM MAE to prevent overfitting to subject-specific physiological characteristics.

V. Deployment & Real-Time Optimization

A. Hugging Face Spaces & Gradio Infrastructure

The VitalEye system is deployed as a cloud-native web application on Hugging Face Spaces using the Gradio framework. This choice provides a highly accessible, interactive interface for researchers and clinicians. The application is containerized using Docker, ensuring environment parity between development and production. The deployment includes automated weight management, where the system fetches the latest YOLOv8 and PhysFormerX checkpoints from the Hugging Face Hub if they are not cached locally, enabling seamless updates and scalability.

B. OpenVINO CPU Acceleration

A critical challenge in deploying deep learning models on shared infrastructure like Hugging Face Spaces is the limited availability of high-end GPUs. To achieve real-time performance on commodity CPU hardware, we integrated OpenVINO (Open Visual Inference & Neural Network Optimization) acceleration. During the initialization phase, the YOLOv8 face-tracking model is dynamically exported and compiled to the OpenVINO format. This optimization leverages Intel-specific instruction sets (AVX-512, VNNI) to achieve a 5x-10x speedup in inference latency compared to standard PyTorch execution on the CPU.

C. Causal Sliding Window Inference

To provide a continuous monitoring experience, the deployment pipeline implements a causal sliding window inference strategy. Rather than processing the entire video at once, the system maintains a 10-second temporal buffer (300 frames) and shifts it with a 1-second stride. This allows for near-real-time updates of Heart Rate and Respiratory Rate as the video progresses. We utilize a Hann window and FFT-based estimation within each sliding window to ensure stability, combined with an Exponential Moving Average (EMA) and OneEuroFilter for bounding box smoothing to prevent jitter-induced noise.

D. Web-Optimized Video Processing

The deployment pipeline is optimized for low-latency web playback. We implemented a batch-processing strategy for YOLOv8 detection (batch size of 16) to maximize throughput. To ensure compatibility with browser-based video players, the final annotated output is programmatically transcoded using FFMPEG with the H.264 (libx264) codec and the 'ultrafast' preset. This ensures that processed results are viewable immediately without requiring additional client-side decoders.

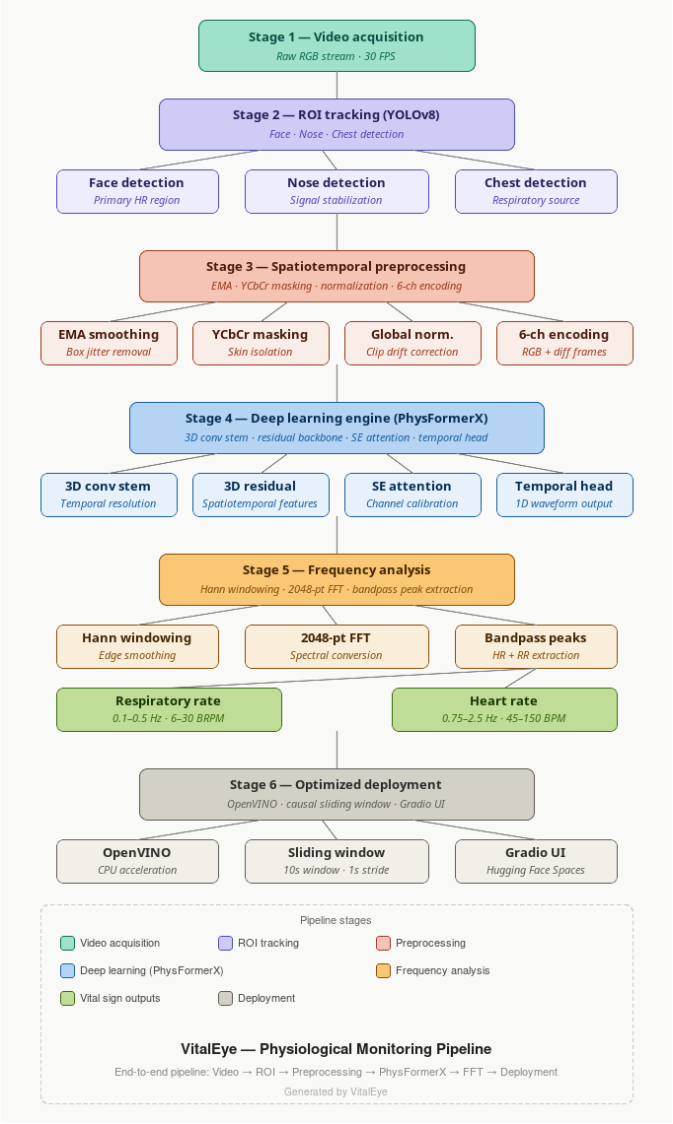

Fig. 2. The VitalEye modular pipeline for remote physiological monitoring, integrating YOLOv8 detection, robust preprocessing, and PhysFormerX-based signal extraction.

VI. Dataset Preparation

The primary dataset used in this project is the UBFC-rPPG dataset. It contains videos of 42 subjects recorded under varied conditions, including resting states and during mathematical tasks that introduce subtle facial movements and physiological variations.

A. Preparation Pipeline

The dataset preparation process involved multiple stages to transform raw videos into a format suitable for training and inference:

- **Raw Videos**: The original dataset consists of high-resolution RGB videos.
- **Video Compression**: All videos were first compressed to reduce storage requirements and improve processing efficiency without significantly affecting signal quality.
- **Bounding Box Data Generation**: The compressed videos were processed using MediaPipe along with a custom mathematical model to generate bounding box coordinates for regions of interest (ROIs). These ROIs include facial and upper-body regions relevant for physiological signal extraction.
- **YOLO Training Dataset Creation**: The processed data was structured in the following format:
  - A folder containing extracted video frames
  - A corresponding CSV file containing bounding box coordinates for each frame
  - All data organized under a folder named after each video

  This structure was used to train the YOLOv8 model for automated ROI detection.
- **YOLO-based ROI Extraction (Final Dataset Creation)**: After training, the YOLO model was applied to the original (uncompressed) videos to generate the final dataset. The output was organized as:

```
database/
  |-- video_name/
      |-- face.avi
      |-- nose.avi
      |-- chest.avi
      |-- ground_truth.txt
```

  Each folder contains ROI-specific video clips along with synchronized physiological ground truth signals.

B. Final Dataset Structure

Each video has its own directory containing:

- ROI-specific videos (Face, Nose, Chest)
- Corresponding ground truth physiological signals

This structured format ensures clean data handling and supports efficient model training and evaluation.

VII. Experimental Depth and Tried Methods

A key strength of this project was the broad experimentation. We did not just build one model; we explored and benchmarked several architectures to find the most robust solution.

A. DeepPhys: Convolutional Attention Network (CAN) Architecture

The CAN architecture utilizes a dual-stream approach to jointly learn ROI detection and physiological signal estimation. **Motion Model**: A VGG-style CNN that estimates the signal derivative $p'(t)$. We use Average Pooling because combining features in physiological sensing often yields a higher SNR. Tanh is used for activation as its symmetry benefits signal estimation, while the gating mechanism prevents vanishing gradients. **Appearance Model**: A parallel stream processing raw frames to generate spatial-temporal attention masks. **Novel Motion Representation: Normalized Frame Difference**: DeepPhys utilizes a "Normalized Frame Difference" as the input for the motion model. The derivation of our input representation $D_l(t)$ involves two key steps: 1) Spatial Averaging: Frames are downsampled to $L \times L$ (typically 36×36) via bicubic interpolation to reduce camera quantization noise $v_n(t)$. 2) Temporal Derivative and Normalization: Taking the first-order derivative removes the dominant stationary skin reflection. We normalize it by dividing it by the temporal mean of $C_l(t)$ to eliminate the stationary luminance. The resulting discrete expression for the motion input is: $D_l(t) \approx \frac{C_l(t+\Delta t)+C_l(t)}{C_l(t+\Delta t)-C_l(t)}$.

B. PhysNet

PhysNet3DCNN-ED is a 3D convolutional encoder-decoder network designed to extract remote photoplethysmography (rPPG) signals from video frames. Input takes a video clip of shape (B, 3, T, H, W). Encoder progressively compresses the input through three stages. A stem conv with $1 \times 5 \times 5$ filters captures spatial appearance first. Then two deeper blocks of four 3D convolutions each learn richer features. Critically, the 2nd and 3rd MaxPool layers use $2 \times 2 \times 2$ kernels, which downsample temporally as well as spatially, squeezing the signal to $T/4$ at the bottleneck. The decoder uses two ConvTranspose3d layers with $2 \times 1 \times 1$ kernels to upsample only the temporal dimension back to the original T. The head applies global average pooling over H and W, then a $1 \times 1 \times 1$ conv reduces channels to 1, yielding the predicted rPPG waveform.

C. HybridRPPG

It combines two complementary signal extraction strategies: **CAN spatial branch** processes each frame independently using 2D convolutions with learned attention masks to locate high-quality skin regions and outputting a per-frame scalar signal. **PhysNet-ED temporal branch** sees the video clip as a 3D volume. Its encoder downsamples temporally to capture long-range pulse rhythms. Crucially, the motion clip is multiplied by the CAN attention volume so PhysNet only processes skin-focused regions. **Fusion gate** stacks both outputs and runs them through a small Conv1d network that learns per-timestep blend weights. The final output is the weighted sum of both branches.

D. TempResNet (Temporal Resolution Network)

A pure 3D CNN architecture engineered to preserve the time dimension throughout feature extraction. **Spatial-Only Pooling**: Uses MaxPool3d((1, 2, 2)) to compress spatial data while maintaining a temporal stride of 1. **1D Temporal Regression**: A Conv1d regressor maps abstract features directly

into a continuous 1D physiological waveform. **Phase-Driven Optimization**: Optimized with a Negative Pearson correlation loss. **Performance**: Achieved a Mean Absolute Error (MAE) of 12.71 BPM.

E. AdvancedPhysNet

An upgraded 3D ResNet architecture that adds channel-wise attention and temporal smoothing. **Spatial-Thinned Res-Blocks**: Uses 3D Residual Blocks with asymmetric strides (1, 2, 2). **Channel Attention & Smoothing**: Integrates Squeeze-and-Excitation (SE) modules and a 1D Temporal Convolution. **Hybrid Loss & Stable Tracking**: Trained with a combined Negative Pearson and Smooth L1 loss, supported by YOLO-based face tracking. **Performance**: Achieved an MAE of 8.23 BPM.

F. Signal Enhancement Techniques

In addition to deep learning models, we explored several signal enhancement techniques to improve robustness.

1) Eulerian Video Magnification (EVM): EVM is a signal enhancement technique used to amplify subtle color and motion variations in video that are otherwise invisible to the human eye. It works by first isolating relevant spatial components of the video and then applying a temporal bandpass filter to focus on physiological frequency ranges such as heart rate (≈0.9–2 Hz) and respiratory rate (≈0.1–0.5 Hz). The filtered signal is then amplified by a factor (α) and reconstructed back into video frames, making small physiological changes more visible and measurable.

The main purpose of EVM in the system is to enhance weak signals such as blood flow-induced skin color changes and breathing motion, thereby improving signal quality before further processing. However, an important limitation is that EVM amplifies all variations, not just useful physiological signals. This includes noise from lighting changes, camera artifacts, and motion. Due to this drawback, EVM must be used carefully and is typically applied selectively depending on signal quality (SNR). In practical scenarios, especially with motion and uncontrolled environments, excessive amplification can degrade performance rather than improve it, which is why it is not always included in the final pipeline.

2) Chrominance-Based rPPG: This is a mathematical approach that extracts the blood volume pulse by first normalizing facial RGB signals to eliminate general illumination intensity variations. The process transforms these signals into two linear combinations, X and Y, specifically designed to preserve pulse-related color changes while suppressing common-mode motion artifacts. These components are band-pass filtered, and an adaptive scaling factor α is applied to balance them, resulting in the final pulse signal $S = X - \alpha Y$. This mathematical projection isolates subtle blood-volume-induced variations from significant noise. It is a non-learning baseline that we use as a backup for integrating with the models. The test MAE came out at around 27.11 BPM when integrated with PhysFormerX.

3) Lightweight CNN: This method presents a lightweight, non-end-to-end machine learning pipeline for real-time heart rate (HR) estimation from standard video cameras, specifically designed to handle realistic challenges like head movement and varying illumination while remaining highly interpretable. The system uses the Mediapipe framework to robustly track 3D facial regions of interest and extracts the raw remote photoplethysmography (rPPG) signal primarily from the green color channel. This signal is then decomposed via Empirical Mode Decomposition (EMD) into five distinct frequency channels (IMFs), which are fed into a highly efficient 1D Convolutional Neural Network (CNN) equipped with a channel-wise attention mechanism. This attention module not only performs the HR inference but also dynamically assigns visible weights to the specific IMFs containing the actual heartbeat, effectively "opening the black box" to show exactly how the network reaches its conclusions. Combined with a novel temporal peak-merging post-processing step to smooth out signal artifacts, the resulting system achieves high accuracy while requiring minimal storage and running in under 100 milliseconds on a standard laptop CPU without any GPU acceleration. The test MAE was 12.02 BPM and Pearson Correlation was 0.26.

VIII. Results

We conducted a rigorous performance comparison of all implemented models on the UBFC-RPPG test set. The results are summarized in Table I.

TABLE I
PERFORMANCE COMPARISON OF HEART RATE ESTIMATION MODELS ON UBFC-RPPG TEST SET

Model	Test MAE (BPM)
DeepPhys (MSE Loss)	26.448
PhysNet (Negative Pearson Correlation Loss)	44.992
HybridRPPG (Hybrid Loss)	18.642
HybridRPPG + EVM Videos (Hybrid Loss)	20.888
VidFormer + Chrominance Features	27.11
Lightweight CNN	12.02
PhysformerX + Advanced PhysNet + Preprocessing	6.28
Vidformer + Preprocessing	9.632
Advanced PhysNet + Preprocessing	8.23

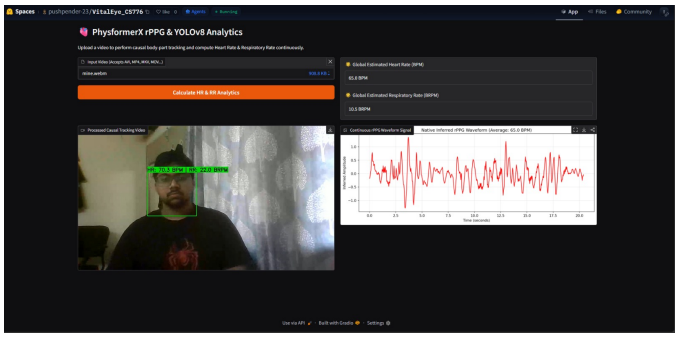

Fig. 3. The VitalEye web interface deployed on Hugging Face Spaces, featuring real-time video upload, ROI visualization, and synchronized heart rate and respiratory rate waveform plotting.

Key Insight: The results clearly demonstrate that architecture alone is not enough. PhysNet, despite being a powerful 3D CNN, failed completely when applied to raw data. Our final pipeline reached an MAE of 6.28 BPM because it combined advanced architecture (PhysFormerX) with engineering-heavy preprocessing (ROI stabilization and 6-channel difference frames). This synergistic approach is what makes VitalEye robust.

IX. Conclusion and Future Work

The VitalEye project demonstrates that a carefully designed multi-stage pipeline can effectively address the core challenges of remote photoplethysmography (rPPG), including motion artifacts and illumination variability. By integrating robust detection (YOLOv8), advanced spatiotemporal preprocessing, and deep learning architectures (PhysFormerX), the system achieves strong performance with a Mean Absolute Error (MAE) of 6.28, highlighting its practical viability.

This work establishes that contactless vital sign monitoring using standard RGB video is not only feasible but reliable when combining deep learning with principled signal processing techniques. The success of the pipeline underscores the importance of each stage, particularly preprocessing, in enhancing signal quality and ensuring model robustness.

A. Key Takeaways

- The integration of deep learning and signal processing significantly improves rPPG performance.
- Preprocessing is critical—stabilization and normalization directly impact final accuracy.
- 6-channel input representations enhance signal-to-noise ratio and model sensitivity to physiological signals.

B. Future Work

Future efforts will focus on extending the system for multi-person tracking and deploying it in real-world clinical environments, such as hospital wards. Additionally, we aim to explore advanced applications including blood pressure estimation and affective state analysis, further expanding the scope of contactless physiological monitoring.

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